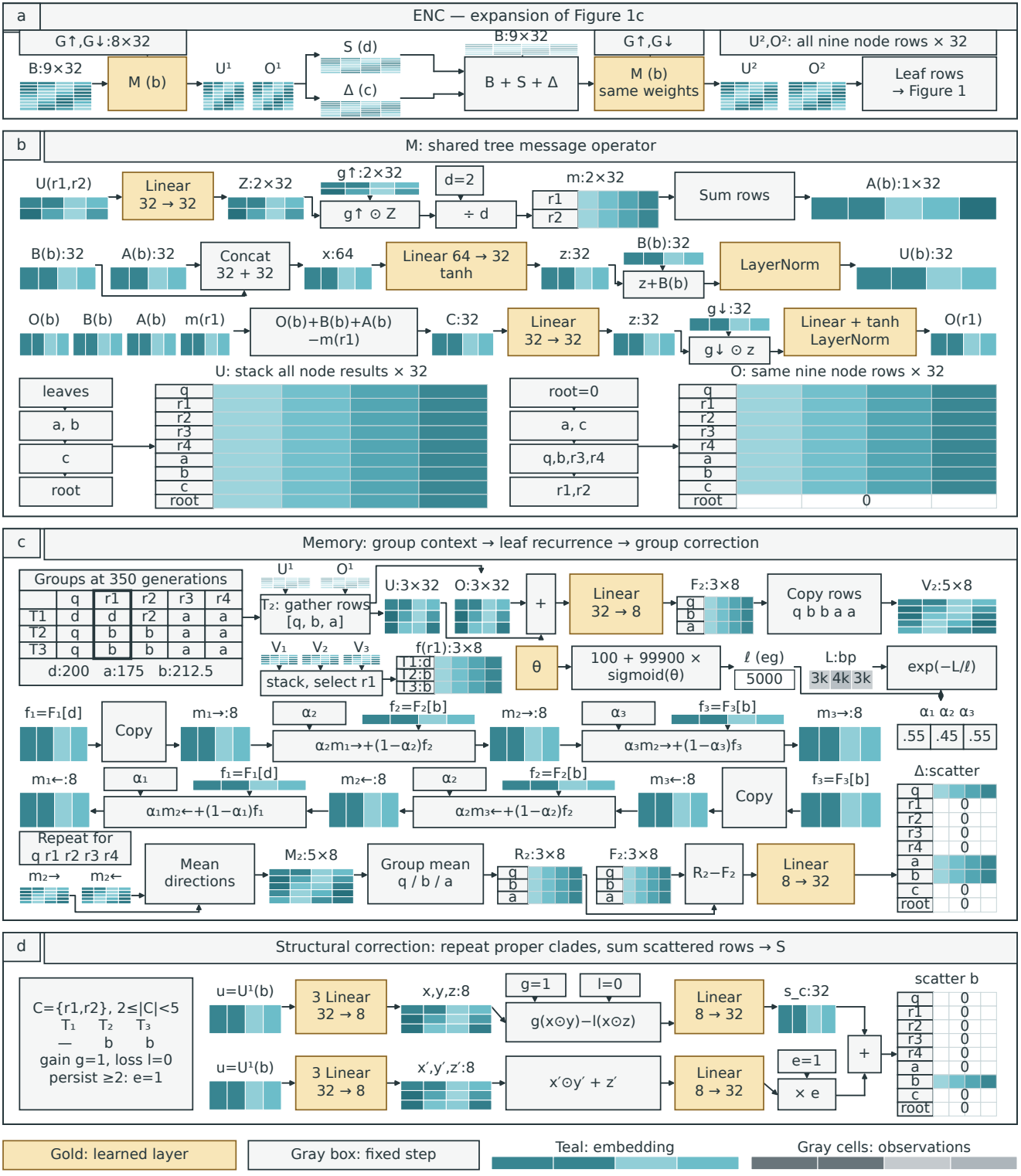
a
ENC — expansion of Figure 1c
G↑,G↓:8×32
B:9×32
M (b)
U¹
O¹
S (d)
Δ (c)
B:9×32
B + S + Δ
G↑,G↓
M (b)
same weights
U²,O²: all nine node rows × 32
U²
O²
Leaf rows
→ Figure 1
b
M: shared tree message operator
U(r1,r2)
Linear
32 → 32
Z:2×32
g↑:2×32
g↑ ⊙ Z
d=2
÷ d
m:2×32
r1
r2
Sum rows
A(b):1×32
B(b):32
A(b):32
Concat
32 + 32
x:64
Linear 64 → 32
tanh
z:32
B(b):32
z+B(b)
LayerNorm
U(b):32
O(b)
B(b)
A(b)
m(r1)
O(b)+B(b)+A(b)
−m(r1)
C:32
Linear
32 → 32
z:32
g↓:32
g↓ ⊙ z
Linear + tanh
LayerNorm
O(r1)
leaves
a, b
c
root
U: stack all node results × 32
q
r1
r2
r3
r4
a
b
c
root
root=0
a, c
q,b,r3,r4
r1,r2
O: same nine node rows × 32
0
c
Memory: group context → leaf recurrence → group correction
Groups at 350 generations
q
r1
r2
r3
r4
T1
d
d
r2
a
a
T2
q
b
b
a
a
T3
q
b
b
a
a
d:200 a:175 b:212.5
U¹
O¹
T₂: gather rows
[q, b, a]
U:3×32
O:3×32
+
Linear
32 → 8
F₂:3×8
g
b
a
Copy rows
q b b a a
V₂:5×8
V₁
V₂
V₃
stack, select r1
f(r1):3×8
T1:d
T2:b
T3:b
θ
100 + 99900 ×
sigmoid(θ)
ℓ (eg)
5000
L:bp
3k 4k 3k
exp(−L/ℓ)
f₁=F₁[d]
Copy
m₁→:8
α₂
f₂=F₂[b]
α₂m₁→+(1−α₂)f₂
m₂→:8
α₃
f₃=F₃[b]
α₃m₂→+(1−α₃)f₃
m₃→:8
α₁ α₂ α₃
.55 .45 .55
m₁←:8
α₁
f₁=F₁[d]
α₁m₂←+(1−α₁)f₁
m₂←:8
α₂
f₂=F₂[b]
α₂m₃←+(1−α₂)f₂
m₃←:8
Copy
f₃=F₃[b]
Δ:scatter
q
r1
r2
r3
r4
a
b
c
root
0
Repeat for
q r1 r2 r3 r4
m₂→
m₂←
Mean
directions
M₂:5×8
Group mean
q / b / a
R₂:3×8
F₂:3×8
R₂−F₂
Linear
8 → 32
d
Structural correction: repeat proper clades, sum scattered rows → S
C={r1,r2}, 2≤|C|<5
T₁ T₂ T₃
— b b
gain g=1, loss l=0
persist ≥2: e=1
u=U¹(b)
3 Linear
32 → 8
x,y,z:8
g=1
l=0
g(x⊙y)−l(x⊙z)
Linear
8 → 32
s_c:32
scatter b
u=U¹(b)
3 Linear
32 → 8
x′,y′,z′:8
x′⊙y′ + z′
Linear
8 → 32
e=1
× e
+
Gold: learned layer
Gray box: fixed step
Teal: embedding
Gray cells: observations